

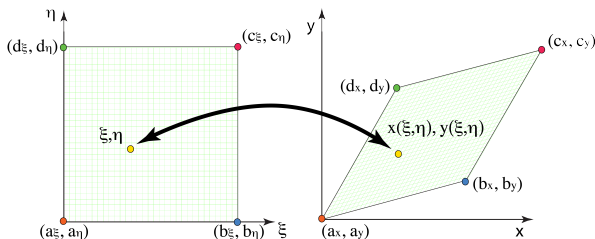
MECH5200 Numerical Methods for PDEs : *Video 13:* *2D Finite Difference Mappings*

David Willis & Xinfang Jin

February 13, 2026

Idea and References

- We'd like to use Finite Differences to solve the PDE on a non-square/rectangular domain
- We've identified that a **mapping** will be used:



- The following materials were used in the preparation of this lecture:
 - ① Tannehill, Anderson and Pletcher, *Computational fluid Mechanics and Heat Transfer*.
 - ② MIT Open Courseware 16.920, lecture 4 Notes
- The author of these slides wishes to thank these sources.

2-D Finite Differences on Irregular Domains – Mappings

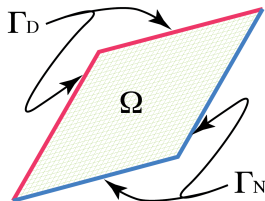
- As an example, let's consider the following problem **in the physical domain** (the process is general and applies to other equations):

$$-\nabla^2 u = f \quad \text{on } \Omega \quad (1)$$

$$u = g \quad \text{on } \Gamma_D \quad (2)$$

$$\frac{\partial u}{\partial n} = h \quad \text{on } \Gamma_N \quad (3)$$

- Assume the values for f , g and h are given.



Aside: Boundary Conditions

- A **Dirichlet** boundary condition prescribes solution value at the boundary.
- A **Neumann** boundary condition prescribes the normal derivative value on the boundary.
- In both cases, only the row of the A-matrix corresponding to the boundary condition is modified!

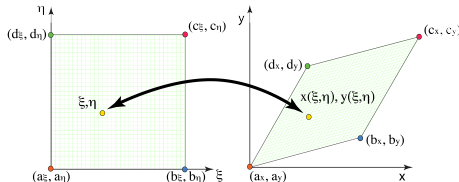
Geometry/Description

- Idea:

- Map the problem from the irregular domain to the regular domain (Video 11 & 12)
- In the process we will create a “new” equation to solve on the computational domain:

$$x = x(\xi, \eta) \quad (4)$$

$$y = y(\xi, \eta) \quad (5)$$



Deriving the Poisson Equation for General Domains

- Let's look at how to go from the **physical** to **computational** domain.

$$\xi = \xi(x, y) \quad \text{and} \quad \eta = \eta(x, y)$$

- Using the total differential for ξ and η we can write:

$$d\xi = \left(\frac{\partial \xi}{\partial x} \right) dx + \left(\frac{\partial \xi}{\partial y} \right) dy$$

$$d\eta = \left(\frac{\partial \eta}{\partial x} \right) dx + \left(\frac{\partial \eta}{\partial y} \right) dy$$

- We can rewrite this as:

$$\begin{bmatrix} d\xi \\ d\eta \end{bmatrix} = \begin{bmatrix} \left\{ \frac{\partial \xi}{\partial x} \right\} & \left\{ \frac{\partial \xi}{\partial y} \right\} \\ \left\{ \frac{\partial \eta}{\partial x} \right\} & \left\{ \frac{\partial \eta}{\partial y} \right\} \end{bmatrix} \begin{bmatrix} dx \\ dy \end{bmatrix}$$

Deriving the Poisson Equation for General Domains

- Let's look at how to go from the **computational** to **physical** domain.

$$x = x(\xi, \eta) \quad \text{and} \quad y = y(\xi, \eta)$$

- Starting with the differential of ξ and η :

$$dx = \left(\frac{\partial x}{\partial \xi} \right) d\xi + \left(\frac{\partial x}{\partial \eta} \right) d\eta$$

$$dy = \left(\frac{\partial y}{\partial \xi} \right) d\xi + \left(\frac{\partial y}{\partial \eta} \right) d\eta$$

- We can rewrite this as:

$$\begin{bmatrix} dx \\ dy \end{bmatrix} = \begin{bmatrix} \left(\frac{\partial x}{\partial \xi} \right) & \left(\frac{\partial x}{\partial \eta} \right) \\ \left(\frac{\partial y}{\partial \xi} \right) & \left(\frac{\partial y}{\partial \eta} \right) \end{bmatrix} \begin{bmatrix} d\xi \\ d\eta \end{bmatrix} \rightarrow \begin{bmatrix} d\xi \\ d\eta \end{bmatrix} = \begin{bmatrix} \left(\frac{\partial x}{\partial \xi} \right) & \left(\frac{\partial x}{\partial \eta} \right) \\ \left(\frac{\partial y}{\partial \xi} \right) & \left(\frac{\partial y}{\partial \eta} \right) \end{bmatrix}^{-1} \begin{bmatrix} dx \\ dy \end{bmatrix}$$

Deriving the Poisson Equation for General Domains

- So we can see from this that the mapping derivatives are related as:

$$\begin{bmatrix} \left(\frac{\partial \xi}{\partial x} \right) & \left(\frac{\partial \xi}{\partial y} \right) \\ \left(\frac{\partial \eta}{\partial x} \right) & \left(\frac{\partial \eta}{\partial y} \right) \end{bmatrix} = \begin{bmatrix} \left(\frac{\partial x}{\partial \xi} \right) & \left(\frac{\partial x}{\partial \eta} \right) \\ \left(\frac{\partial y}{\partial \xi} \right) & \left(\frac{\partial y}{\partial \eta} \right) \end{bmatrix}^{-1} = \frac{1}{J} \begin{bmatrix} \left(\frac{\partial y}{\partial \eta} \right) & - \left(\frac{\partial x}{\partial \eta} \right) \\ - \left(\frac{\partial y}{\partial \xi} \right) & \left(\frac{\partial x}{\partial \xi} \right) \end{bmatrix}$$

- This allows us to compute the mapping derivatives in **the computational domain**.

$$\begin{aligned} \left(\frac{\partial \xi}{\partial x} \right) &= \frac{1}{J} \left(\frac{\partial y}{\partial \eta} \right) & \text{and} & & \left(\frac{\partial \xi}{\partial y} \right) &= \frac{-1}{J} \left(\frac{\partial x}{\partial \eta} \right) \\ \left(\frac{\partial \eta}{\partial x} \right) &= \frac{-1}{J} \left(\frac{\partial y}{\partial \xi} \right) & \text{and} & & \left(\frac{\partial \eta}{\partial y} \right) &= \frac{1}{J} \left(\frac{\partial x}{\partial \xi} \right) \end{aligned}$$

Deriving the Poisson Equation for General Domains

- We want to determine u on the **computational** domain by solving an equivalent equation:

$$u(x, y) = u(x(\xi, \eta), y(\xi, \eta))$$

- Start by taking the first derivative w.r.t. x and y using the chain rule (recall, we'll need second derivatives):

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial x} \\ \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial y}\end{aligned}$$

- Switch the terms $\frac{\partial \xi}{\partial x}$, $\frac{\partial \xi}{\partial y}$, $\frac{\partial \eta}{\partial x}$ and $\frac{\partial \eta}{\partial y}$ with equivalent derivatives w.r.t. the **computational** domain variables.

Deriving the Poisson Equation for General Domains

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial x} \\ \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial y}\end{aligned}$$

- Substituting the terms from earlier:

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{1}{J} \frac{\partial u}{\partial \xi} \left(\frac{\partial y}{\partial \eta} \right) + \frac{-1}{J} \frac{\partial u}{\partial \eta} \left(\frac{\partial y}{\partial \xi} \right) \\ \frac{\partial u}{\partial y} &= \frac{-1}{J} \frac{\partial u}{\partial \xi} \left(\frac{\partial x}{\partial \eta} \right) + \frac{1}{J} \frac{\partial u}{\partial \eta} \left(\frac{\partial x}{\partial \xi} \right)\end{aligned}$$

- The process is repeated now to find the second derivative terms with respect to the **computational** domain variables.
- See the MIT 16.290 Open courseware notes for details.

Deriving the Poisson Equation for General Domains

- When we follow the same process for the second derivatives, $\frac{\partial^2 u}{\partial x^2}$ and $\frac{\partial^2 u}{\partial y^2}$, the Laplace operator:

$$-\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = -(u_{xx} + u_{yy}) = f$$

- Can be written as:

$$\begin{aligned} \frac{-1}{J^2} \left(a \frac{\partial^2 u}{\partial \xi^2} - 2b \frac{\partial^2 u}{\partial \xi \partial \eta} + c \frac{\partial^2 u}{\partial \eta^2} + d \frac{\partial u}{\partial \eta} + e \frac{\partial u}{\partial \xi} \right) &= f \\ \frac{-1}{J^2} (a u_{\xi\xi} - 2b u_{\xi\eta} + c u_{\eta\eta} + d u_{\eta} + e u_{\xi}) &= f \end{aligned}$$

Deriving the Poisson Equation for General Domains

- Where, a, b, c, d , and e depend on the mapping as:

$$a = x_\eta^2 + y_\eta^2 \quad e = \frac{x_\eta \beta - y_\eta \alpha}{J} \quad \alpha = ax_{\xi\xi} - 2bx_{\xi\eta} + cx_{\eta\eta} \quad (6)$$

$$b = x_\eta x_\xi + y_\eta y_\xi \quad d = \frac{y_\xi \alpha - x_\xi \beta}{J} \quad \beta = ay_{\xi\xi} - 2by_{\xi\eta} + cy_{\eta\eta} \quad (7)$$

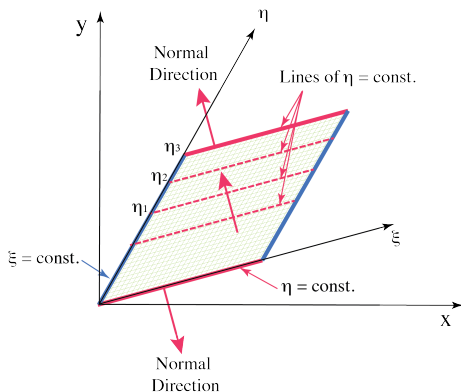
$$c = x_\xi^2 + y_\xi^2 \quad J = x_\xi y_\eta - x_\eta y_\xi \quad (8)$$

Finite Differences in Two Dimensions – Regular Domain

- We have transformed the equation from one domain to another.
- But...we haven't transformed the boundary conditions.
- The Dirichlet boundary condition is the same ($u(x(\xi, \eta), y(\xi, \eta)) = g$)
- But, the Neumann boundary condition is not yet defined for the mapping.

Finite Differences in Two Dimensions – Regular Domain

- Let's look at the normal boundary condition, $\frac{\partial u}{\partial n} = \nabla u \cdot \hat{n}$:



- Lets work on defining $\hat{n}$: The normal vector on the top boundary is in the direction of the gradient of η – i.e. parallel to $\nabla\eta$.

Finite Differences in Two Dimensions – Regular Domain

- The gradient is:

$$\nabla(\eta) = \left[\frac{\partial \eta}{\partial x}, \frac{\partial \eta}{\partial y} \right]$$

- We want the unit normal (vector length = 1), so we must normalize this:

$$\hat{n} = \frac{1}{\sqrt{\left(\frac{\partial \eta}{\partial x}\right)^2 + \left(\frac{\partial \eta}{\partial y}\right)^2}} \left[\frac{\partial \eta}{\partial x}, \frac{\partial \eta}{\partial y} \right]$$

- But, we want derivatives in terms of the computational domain, so we substitute our earlier mapping derivative equivalencies (J's cancel):

$$\hat{n} = \frac{1}{\sqrt{\left(\frac{\partial x}{\partial \xi}\right)^2 + \left(\frac{\partial y}{\partial \xi}\right)^2}} \left[-\frac{\partial y}{\partial \xi}, \frac{\partial x}{\partial \xi} \right]$$

Finite Differences in Two Dimensions – Regular Domain

- So, the normal vector for the $\eta = \text{const}$ edge is $\hat{n}$:

$$(n^x, n^y) = \frac{1}{\sqrt{\left(\frac{\partial x}{\partial \xi}\right)^2 + \left(\frac{\partial y}{\partial \xi}\right)^2}} \left[-\frac{\partial y}{\partial \xi}, \frac{\partial x}{\partial \xi} \right] \quad (9)$$

- The gradient of the unknown, u is:

$$\begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{1}{J} \frac{\partial u}{\partial \xi} \left(\frac{\partial y}{\partial \eta} \right) + \frac{-1}{J} \frac{\partial u}{\partial \eta} \left(\frac{\partial y}{\partial \xi} \right) \\ \frac{-1}{J} \frac{\partial u}{\partial \xi} \left(\frac{\partial x}{\partial \eta} \right) + \frac{1}{J} \frac{\partial u}{\partial \eta} \left(\frac{\partial x}{\partial \xi} \right) \end{bmatrix}$$

- The Neumann boundary condition for the domain edge where $\eta = \text{const}$ is:

$$\begin{aligned} \frac{\partial u}{\partial n} &= \nabla u \cdot \hat{n} = \frac{\partial u}{\partial x} n^x + \frac{\partial u}{\partial y} n^y \\ &= \frac{1}{J} \left[\left(\frac{\partial y}{\partial \eta} n^x - \frac{\partial x}{\partial \eta} n^y \right) \frac{\partial u}{\partial \xi} + \left(-\frac{\partial y}{\partial \xi} n^x + \frac{\partial x}{\partial \xi} n^y \right) \frac{\partial u}{\partial \eta} \right] \end{aligned}$$

What have we learned?

- When solving the problem in the computational domain, we must modify the equation.
- The derivation of the **computational** domain equation is:
 - Similar for 1-D and 2-D (look for derivatives w.r.t. computational domain variables, ξ, η).
 - Unique for each different governing equation (need to map equation of interest).
 - Complex, but creates a relationship between the **physical** and **computational** domain.
 - Dirichlet (value assigned) boundary conditions are simple.
 - Neumann (normal derivative) boundary conditions are more challenging.